

Alps Alpine Dual Switching Magnetic Sensor Board (Qwiic) - Application Notes

These are the application notes for the Alps Alpine Dual Switching Magnetic Sensor Board (Qwiic).

I2C Address Assignments

A0	A1	I2C Address
0	0	0x20 (Default)
0	1	0x21
1	0	0x22
1	1	0x23

Message Sequence & Format

All Alps Alpine QWIIC Sensor/HMI boards use a similar I²C communication sequence between the master and slave. In these exchanges, the Alps Alpine QWIIC Sensor/HMI boards always functions as the slave device.

The master is responsible for initiating both read and write operations. To change a setting or request data from a board, follow the sequence below:

**** Placeholder Sequence. Add diagram later ****

- The Master must first send request message → The board will prepare the data or change setting as requested
- The Master must then send read message with size required for the response → The board will send back the requested data response

Request Message Types

The following message types are used to indicate whether the board should prepare requested data for the user to read back or execute an operation. This message type ID is located at the beginning of the message payload sent from the I2C master device to the Alps Alpine Sensor/HMI board.

Request type	Message ID Value (uint8)	Description
Request Data	0x01	Used to request the board to prepare data that is in the data request (proceeding value in the request payload)
Request Operation	0x02	Used to request the board to execute a operation (proceeding operation in the request payload)

Request Data Types & Formats (0x01):

The following are different request data types that are supported by the Alps Alpine Dual Switching Magnetic Sensor Board (Qwiic) and their expected formats.

“WHO AM I” Data Request Message (0x01):

The “WHO AM I” message can be used to both determine if a Alps Alpine Sensor/HMI board is present and can be used to determine what type of Sensor/HMI board it is.

Request I2C Payload:

Byte Index	Value	Description
0	0x01	Request Data type
1	0x01	Who Am I Data Request

Response I2C Payload:

During response request for “WHO AM I”, Master must perform a read request with **3 bytes**.

Byte Index	Value	Description
0	0x01	Message Header
1	0x01	Message Header
2	0x0A	Device type indicator. 0x0A = Alps Alpine Dual Switching Magnetic Sensor

		Board (Qwiic)
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“Get Formatted Values” Data Request Message (0x02):

The “Get formatted values” message, can be used to request the current values from the Sensor board in a more friendly format for reading. Response data and its format may change based on current set modes. In the formatted request, the detection of pole is provided as a single byte enum value to present detections.

Request I2C Payload:

Byte Index	Value	Description
0	0x01	Request Data type
1	0x02	Request Formatted Values

Response I2C Payload:

In default operating mode, during the response request for “Get Formatted Values Data”, the Master must perform a read request with **1 byte**. This byte indicates what magnetic pole is currently detected.

Byte Index	Content	Description
0	DetectedMagneticPole 0x00 ~ 0x02 (uint8_t)	0x00 = No Pole Detection 0x01 = South Pole Detected 0x02 = North Pole Detected

“Get Raw Values” Data Request Message (0x03):

The “Get Raw values” message, can be used to request the current raw data that the Sensor board reads from the Alps Alpine component. Response data and its format may change based on current set modes.

Request I2C Payload:

Byte Index	Value	Description
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0	0x01	Request Data type
1	0x03	Request Raw Values

Response I2C Payload:

In default operating mode, during the response request for “Get Data”, Master must perform a read request with **2 bytes**. These two bytes represent raw pin readings from the sensor and based on which pin is read “low” (0x00) will be used to indicate if the sensor is detecting a magnetics South or North pole.

Byte Index	Content	Description
0	South Switch Detection 0x00 ~ 0x01 (uint8_t)	0x00 = South Pole Detection 0x01 = South Pole Not Detected
1	North Switch Detection 0x00 ~ 0x01 (uint8_t)	0x00 = North Pole Detection 0x01 = North Pole Not Detected

“Get Is Data Ready” Data Request Message (0x04):

The “Get Is Data Ready” message can be used to check whether the Sensor board is ready to provide component data readings when requested. It is good practice to perform this request after powering on the board and before requesting any data, continuing until the board indicates that it is ready. This message should also be used after any operation request to wait until the board is ready before sending additional requests.

Request I2C Payload:

Byte Index	Value	Description
0	0x01	Request Data type
1	0x04	Request Is Data Ready Value

Response I2C Payload:

In default operating mode, during the response request for “Get Is Data Ready”, the Master must perform a read request with 1 **byte**.

Byte Index	Content	Description
0	<code>isDataReady</code> <code>(uint8_t)</code>	• 0x00 = Data Not Ready • 0x01 = Data Ready

“Get Firmware Version” Data Request Message (0x05):

The “Get Firmware Version” message can be used to check the firmware version of the Sensor board.

Request I2C Payload:

Byte Index	Value	Description
0	0x01	Request Data type
1	0x05	Request Firmware Version

Response I2C Payload:

In default operating mode, during the response request for “Get Firmware Version”, the Master must perform a read request with **2 bytes**.

Byte Index	Content	Description
0	<code>MAJOR_VERSION</code> <code>(uint8_t)</code>	The firmwares major value
1	<code>MINOR_VERSION</code> <code>(uint8_t)</code>	The firmwares minor value

Request Operation Types & Formats (0x02):

The following are different request operation types that are supported by the Alps Alpine Dual Switching Magnetic Sensor Board (Qwiic) and their expected formats. These requests do not have any responses in return.

“Enter DFU Mode” Operation Request Message (0xff):

The “Enter DFU Mode” message can be used to request the Sensor board to enter its DFU mode. This allows for a external I2C Master device to perform DFU over I2C interface.

Request I2C Payload:

Byte Index	Value	Description
0	0x02	Request Operation type
1	0xff	Request Enter DFU Mode